

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introduction to Computer Architecture
In-Class Exercise (09/10)

1. Fill in the correct sequence of MIPS instructions that implement the following C statement:

a = d + A[c+8];

Assume the variables are assigned to registers as: [\$s1] = a, [\$s2] = b, [\$s3] = c, [\$s4] = d, and the base address of **byte** array A (i.e., &A[0]) is contained in \$s5.

```
addi $t1, _____, 8
add  $t1, _____, $t1
lbu  _____, 0($t1)
add  $s1, _____, $t2
```

Answer:

```
addi $t1,  $s3, 8
add  $t1,  $s5, $t1
lbu  $t2,  0($t1)
add  $s1,  $s4, $t2
```

2. Consider the following MIPS assembly program:

```
Loop:  sll  $t1, $s2, 2
       add $t1, $t1, $s1
       lw  $t0, 0($t1)
       bne $t0, $s3, Exit
       addi $s2, $s2, 1
       j   Loop
Exit:
```

The above assembly is compiled from the following C statement (A is an **integer** array):

while (A[i] == k) i++;

Which registers correspond to:

- Integer variable i?
- Integer variable k?
- Base address of integer array A (&A[0])?

Answer:

- [\$s2] = i
- [\$s3] = k
- [\$s1] = &A[0]

3. The following MIPS assembly code has just executed on your machine:

```
andi $s1, $s1, 20
add  $s1, $s0, $s1
lw   $s3, 0($s1)
```

The current contents of registers and memory immediately **after** executing this code are:

Register	Contents
\$s0	4000
\$s1	???
\$s2	3
\$s3	-5

Memory Address	Contents
4000	-5
4004	1
4008	3
4012	7
4016	-5
4020	-5
4024	-1
4028	7
4032	-5
4036	3
4040	3
4044	-1

Given the current state of registers and memory, list all the possible values of [\$s1].

Answer:

4000, 4016, 4020. Why? The ‘andi’ forces all bits to zero except for the 2 bits in 0x14 (n0n00). Because of that, the only numbers that could be added to 4000 in instruction 2 is: 00000, 00100, 10000, 10100 – aka 0, 4, 16, 20, and when added to 4000 only leave those 3 addresses that contain -5.